

AS stoichiometry homework 3 · empirical and molecular formulae

CIE 9701 AS Chemistry · 18 MCQ · 18 marks

Name: _____ Class: _____ Date: _____

Topics: 8 empirical formulae · 9 molecular formulae · 10 hydrates · 11 combustion analysis · 12 formulae and equations.
Later questions also use earlier skills.

Circle **one** letter (A–D) for each question. Show working. Calculators and a Periodic Table may be used.

8 · Empirical formulae

1. What is the empirical formula of butanoic acid?

- A $\text{C}_2\text{H}_4\text{O}$ B $\text{C}_3\text{H}_6\text{O}$
C $\text{C}_4\text{H}_8\text{O}$ D $\text{C}_5\text{H}_{10}\text{O}$

[1]

2. Compound X consists of 40.0% carbon, 6.7% hydrogen and 53.3% oxygen by mass.

What is the empirical formula of compound X?

- A CH_2O B $\text{C}_2\text{H}_2\text{O}$
C $\text{C}_2\text{H}_4\text{O}$ D CHO

[1]

3. Compound X is an organic compound that contains 30.6% carbon, 3.8% hydrogen, 20.4% oxygen and 45.2% chlorine by mass.

What is the empirical formula of X?

- A $\text{C}_2\text{H}_3\text{OCl}$ B $\text{C}_2\text{H}_4\text{OCl}$
C $\text{C}_3\text{H}_4\text{OCl}$ D $\text{C}_4\text{H}_3\text{O}_2\text{Cl}_2$

[1]

4. R is an oxide of Period 3 element T. 5.00 g of R contains 2.50 g of T.

What is T?

- A. magnesium
B. aluminium
C. silicon
D. sulfur

[1]

9 · Molecular formulae

5. Compound X is a straight chain hydrocarbon with an M_r of 84.

What can be determined about X?

1. empirical formula
2. molecular formula
3. whether X contains a C=C bond or not

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

[1]

6. An aqueous solution contains 4.00 g of a carboxylic acid, Q. When this solution reacts with an excess of magnesium, 380 cm^3 of gas is produced, measured at s.t.p.

What is the relative formula mass of Q?

- A 59 B 118
C 126 D 236

[1]

7. P is a compound that burns in an excess of oxygen to give carbon dioxide and water only.

2.20 g of P contains 1.20 g of carbon and 0.20 g of hydrogen.

When P is added to a solution of sodium carbonate, bubbles of gas are seen.

What is P?

- A. CH_3CHO
B. $\text{CH}_3\text{CO}_2\text{H}$
C. $\text{CH}_3\text{COCH}_2\text{CH}_2\text{OH}$
D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$

[1]

10 · Hydrated and anhydrous compounds

8. A sample of 35.6 g of hydrated sodium carbonate contains 25.84% sodium ions by mass.

When this sample is heated, anhydrous sodium carbonate and water are formed.

Which mass of water is given off?

- A** 7.2 g **B** 10.6 g
C 14.4 g **D** 21.2 g

$$[1]$$

9. Hydrated cobalt(II) sulfate loses water when heated to give anhydrous cobalt(II) sulfate. All the water of crystallisation is lost to the atmosphere as steam.

When 3.10 g of hydrated cobalt(II) sulfate, $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$, is heated to constant mass the **loss** in mass is 1.39 g.

What is the value of x , to the nearest whole number?

- A** 4 **B** 6
C 7 **D** 11

$$[1]$$

10. An experiment is carried out to determine the value of x in hydrated lithium hydroxide, $\text{LiOH} \cdot x\text{H}_2\text{O}$. A sample of the solid is heated in a crucible over a Bunsen flame.

Complete dehydration takes place; decomposition does **not** occur.

mass of empty crucible / g = Q

mass of crucible with $\text{LiOH} \cdot x\text{H}_2\text{O}$ / g = R

mass of crucible and residue after heating / g = S

Which equation gives the correct value of x ?

- A. $\frac{S - R}{18}$
 B. $\frac{R - S}{18}$
 C. $\frac{23.9(R - S)}{18(S - Q)}$
 D. $\frac{23.9(S - R)}{18(S - Q)}$

$$[1]$$

11 · Combustion analysis

11. Compound X contains the elements C, H and O only.

2.00 g of X produces 4.00 g of carbon dioxide and 1.63 g of water when completely combusted.

What is the empirical formula of X?

- A CHO_2
C $\text{C}_2\text{H}_4\text{O}$
- B $\text{C}_2\text{H}_2\text{O}$
D CH_2O_2

$$[1]$$

12. Substance Q is a hydrocarbon. When 1.00 g of Q is completely burned, 3.22 g of carbon dioxide is produced.

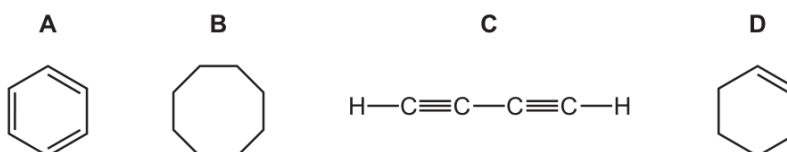
What could be the identity of Q?

- A. cyclohexene
B. cyclopentane
C. ethene
D. pentane

[1]

13. When a small sample of hydrocarbon Q is completely combusted, it produces 3.52 g of carbon dioxide and 1.44 g of water.

What could be the structure of hydrocarbon Q?


$$[1]$$

12 · Chemical formulae and equations

14. Which pair of formulae is correct?

- A. Ag_2CO_3 and $(\text{NH}_4)_3\text{NO}_3$
 B. K_2HCO_3 and $\text{Zn}_3(\text{PO}_4)_2$
 C. AgHCO_3 and K_3PO_4
 D. ZnCO_3 and $(\text{NH}_4)_2\text{PO}_4$

$$[1]$$

15. Chlorine dioxide, ClO_2 , reacts with aqueous sodium hydroxide to produce water and a mixture of two sodium salts, NaClO_2 and NaClO_3 .

What is the mole ratio of NaClO_2 to NaClO_3 in the product mixture?

- A** 1:2 **B** 3:5
C 1:1 **D** 5:3

$$[1]$$

Earlier skills · combining masses, reacting masses, equations

16. Originally, chemists thought indium oxide had the formula InO . By experiment they showed that 4.8 g of indium combined with 1.0 g of oxygen to produce 5.8 g of indium oxide. The A_r of oxygen was known to be 16.

Which value for the A_r of indium is calculated using these data?

- A** 38 **B** 77
C 115 **D** 154

$$[1]$$

17. A 4.00 g sample of an anhydrous Group 2 metal nitrate, Z, is heated strongly until there is no further change of mass. A solid residue of mass 1.37 g is formed.

Which metal is present in Z?

- A. barium
B. calcium
C. magnesium
D. strontium

$$[1]$$

18. Copper dissolves in dilute nitric acid producing a blue solution of $\text{Cu}(\text{NO}_3)_2$, water and nitrogen(II) oxide as the only products.

How many moles of acid react with three moles of copper in the balanced equation?

- | | | | |
|----------|---|----------|---|
| A | 2 | B | 4 |
| C | 6 | D | 8 |

$$[1]$$